

Bronchiolitis - case presentation

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Sudhir Upadhyay, a 5-month old male child, residing in Dharavi, hailing from Solapur, presented to the OPD with chief complaints of cough since 5 days, fever since 3 days and rapid breathing since 1 day.

Informant was mother (reliable)

Date of admission: 12/03/2020

The patient was apparently alright 5 days back, when he developed sudden onset cough which has gradually increased. The mother reports a wet sounding cough in the last 3 days. Since yesterday, the child has woken up in the night 2-3 times due to cough. There are no aggravating factors. Cough is relieved after taking medication.

The child has fever since 3 days. It was sudden in onset, moderate grade, with no diurnal variation. It is not associated with rigors. It is not associated with rashes. Child is normal during inter febrile period. It is relieved with medication.

The child also has a history of fast breathing since 1 day. It is insidious in onset, progressive in nature. Mother reports the presence of chest retractions and wheezing. There is history of irritability and excessive crying since 1 day.

There is no h/o ear pain, ear discharge

No h/o snoring, change in voice

No h/o bluish discoloration of lips or extremities

No h/o vomiting, diarrhea

No h/o altered sensorium/ convulsions

No h/o similar episode in the past

No h/o foreign body ingestion

No h/o nebulisation/ cough more in the night/ other allergies

No h/o contact with TB patient

No past h/o congenital heart disease

Past history

The child did not have any such episode in the past.

There is no h/o bronchial asthma, congenital heart disease or any other illnesses.

Treatment history

Paracetamol and iv fluids were given

The child's symptoms have now improved.

Birth history

Antenatal history

- Registered pregnancy
- All 3 scans were done, IFA and 2 TDAP injections were taken
- No h/o fever during pregnancy
- No h/o diabetes mellitus, hypertension or any other chronic illness in mother

Natal history

- Full term normal vaginal delivery
- Birth weight- 2.8 kg, cried immediately after birth
- No h/o any congenital anomalies
- No h/o NICU admissions

Breastfeeding started within half an hour of delivery. No prelacteal feeds were given.

Postnatal history

- Child is being exclusively breastfed
- There is no h/o any illness or hospitalisation before this

Developmental history

Gross motor: neck holding present, mother says child sometimes sits with support

Fine motor: grasp reflex absent, bidextrous approach

Social skills: identifies mother

Linguistic skills: cooing

The child has achieved developmental milestones as per chronological age.

Immunisation history

At birth: BCG, OPV, Hep B vaccine taken

At 6 weeks: OPV1, Pentavalent 1st dose taken, Rotavirus vaccine, Pneumococcal vaccine, IPV1

At 10 weeks, OPV2, Pentavalent 2nd dose taken, Rotavirus vaccine, Pneumococcal vaccine, IPV2

At 14 weeks, OPV3, Pentavalent 3rd dose taken, Pneumococcal vaccine

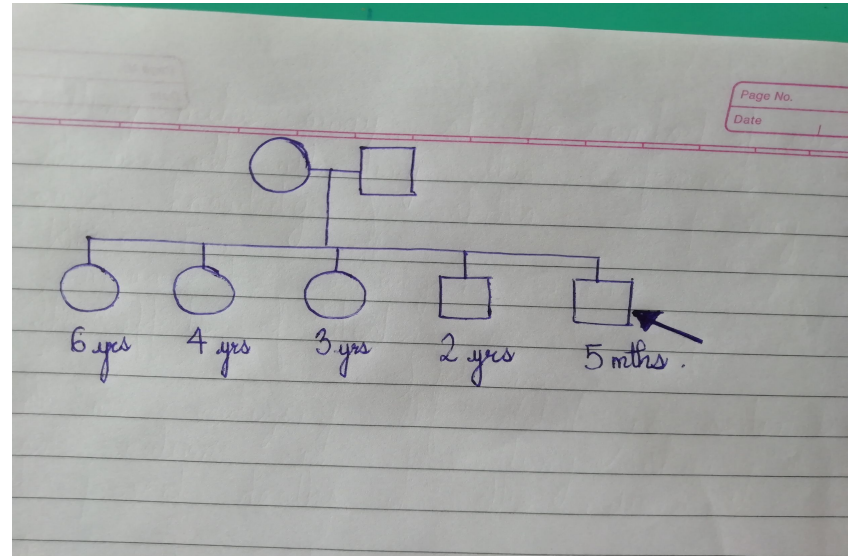
Child is fully immunized upto date according to NIS

Nutritional history

- Breastfeeding was initiated within half an hour of birth. No prelacteal feeds were given.
- Child is exclusively breastfed. He takes regular breastfeeding every 3 hourly for 15-20 minutes (on demand). Night feeds are also being given.
- Child was not accepting feeds and had become irritable in the last 2 days before admission.

Family history

- No h/o genetic or chromosomal disorders in the family
- No familial h/o bronchial asthma/ atopy
- non - consanguinous marriage
- Child is 5th by birth order



Socio-economic history

- Father works as a daily wage labourer and has studied till standard 10th. Mother is a homemaker and has studied till class 8th
- Belongs to upper lower class according to Kuppuswamy classification.
- No h/o firewood use/ smoking in family members
- Overcrowding is present, bad ventilation, no windows and single door in house
- Common toilet in slum area (Sulabh Shauchalaya)
- Intermittent water supply from common tap. Water stored in covered container which is frequently washed.

Summary

A 5 month old male child born out of non- consanguineous marriage presented with chief complaints of cough since 5 days, fever since 3 days and rapid breathing since 1 day. Chest retractions and wheezing are present. There is no h/o any past history of such episode, no family h/o asthma, no suck rest suck cycle and no bluish discoloration. Overcrowding and less birth spacing is present. The child is being exclusively breastfed, completely immunised till date and has normal developmental milestones.

Based on the history, it may be a case of LRTI, probably due to viral etiology.

General examination

- Child is irritable but alert
- Examined in supine position
- Temperature = 100.2 F
- Respiratory rate is 52/min, abdominal, flaring of alae nasi and use of accessory muscles seen
- Pulse : 128 beats per minute, normal in rhythm, volume, tension, character, measured on right radial artery, no radio-radial or radio-femoral delay
- spO2 = 93%

Head to toe examination

- Skull shape: normal, anterior fontanelle open (size: 2.5×2.5 cm), it is not bulging, no pulsations visible. posterior fontanelle closed
- Hair: black, no color changes
- Eyes: no pallor, icterus, KF rings, Bitot's spots, dryness, not sunken
- Ears: no discharge seen
- Nose: flaring of ala nasi seen, no septal deviation
- Lips: no cleft lip, no cyanosis
- Mouth: no cheilosis, stomatitis, bleeding gums
- Tongue: no pallor, cyanosis, oral thrush
- Neck : normal, use of scalene muscles for respiration
- Nails: no clubbing

- Skin: no skin lesions, BCG scar present on left arm
- Abdomen and genitalia: normal
- Spine and back: normal
- Lower limbs: normal
- No peripheral cyanosis
- No edema and lymphadenopathy
- No dysmorphic features

Anthropometry

Weight for age: 8.5 kg ie 85th percentile

Length for age: 62 cm ie 97th percentile

Weight for length: 97th percentile

HC: 44 cm ie 85th percentile

CC: 40 cm

Systemic examination

Upper respiratory tract :

Nose: flaring of ala nasi seen

Oropharynx: appears to be normal, no tonsillar enlargement

Posterior pharyngeal wall: normal

Lower respiratory tract examination

Inspection:

Chest is elliptical in shape

Symmetrical movements on both sides with respiration

Trachea is central in position

Subcostal and intercostal retractions present with use of accessory muscles

Apical beat not seen

No scars, sinuses, dilated veins

No drooping of shoulders, rib crowding

Palpation

- Inspectory findings confirmed
- No tenderness
- No local rise of temperature
- Position of trachea: central
- Apical impulse: felt at 4th intercostal space in MCL
- Chest expansion: equal on both sides
- Vocal fremitus: normal

Percussion

Resonant note elicited on bilateral lung fields

Auscultation

- Normal vesicular sound present on all other lung regions.
- Rhonchi present diffusely on bilateral lung fields
- Vocal resonance : normal

Other systems examination

CVS: no raised JVP, S1 S2 heard

Per Abdomen: No tenderness, no HSM, no ascites

CNS: no apparent sensory or motor deficit

Summary

A 5 month old male child born out of non- consanguineous marriage presented with chief complaints of cough since 5 days, fever since 3 days and rapid breathing since 1 day. Wheezing is also present.

O/E there is presence of fever, intercostal and subcostal retractions present. Diffuse rhonchi present on bilateral lung fields.

Provisional diagnosis

It is a case of LRTI, most probably of viral etiology, most likely acute bronchiolitis

THANK YOU